Naïve Bayes Classifier

Naïve Bayes Classifier -

a. Naive Bayes classifiers are probabilistic models based on Bayes' theorem.

b. It is called naive due to the assumption that the features in the dataset are mutually independent

c. In real world, the independence assumption is often violated, but naive Bayes classifiers still tend to perform very well

d. Idea is to factor all available evidence in form of predictors into the naïve Bayes rule to obtain more accurate probability for class prediction

e. It estimates conditional probability which is the probability that something will happen, *given that something else* has already occurred. For e.g. the given mail is likely a spam given appearance of words such as “prize”

f. Being relatively robust, easy to implement, fast, and accurate, naive Bayes classifiers are used in many different fields

Naïve Bayes Classifier -

Probability - is the number of trials in which an event of interest occurred by total number of trials. (what is a trial and what is an event?)

If it rained 3 out of 10 days in the past where the days were exactly like today, the probability it will rain today is 30%

a. In this example, the day is a trial / experiment
b. The event is rain
c. Probability that it will rain is P(A) (A denoting rains) = 3/10
d. Every trial has at least two outcomes (event will P(A) or will not occur P(A”)
e. The multiple possible events are mutually exclusive i.e. cannot occur simultaneously
f. Total probability in a trial is sum of probabilities of all events = P(A)+P(A”) = 100% = 1

<u>Naïve Bayes Classifier</u> -

Joint Probability – is the probability of multiple events occurring together (we are not talking of causality here i.e. one event leads to another). For e.g.

1. probability of drawing a king from a deck of cards is 4/52
2. Probability of drawing a red colour card from a deck of cards is 26/52
3. Probability of drawing a red colour king = 2 / 52

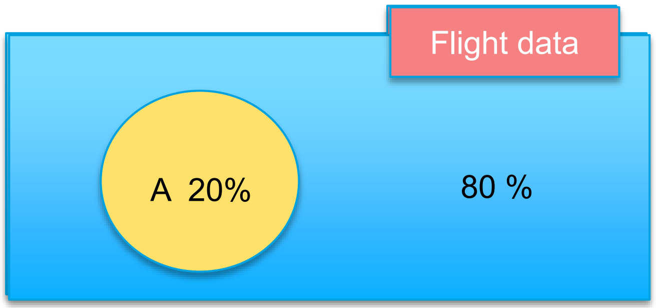
Conditional Probability – it is the probability that an event has occurred (not yet observed) given another event has occurred. For e.g.

1. given the card drawn is red (an event has occurred)
2. what is the probability it is a king (event not yet observed)?
3. Since the card is red, there are 26 likely values for red
4. Of these 26 possible values we are interested in king which is 2 (king of diamonds and heart)
5. Thus the conditional probability that the card is a king given red card is 2 /26
6. Compare this with joint probability of red king (2/52).
7. Given an event has occurred, it increases the probability of the other event

Naïve Bayes Classifier -

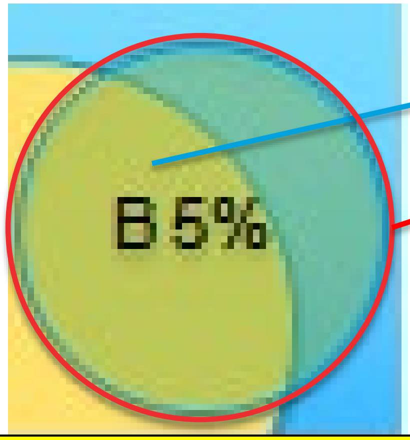
Joint Probabilities -

a. Imagine you represent all the flight experience you had till date as the blue area in a mathematical space. The dimensions of the boxes and circles are immaterial



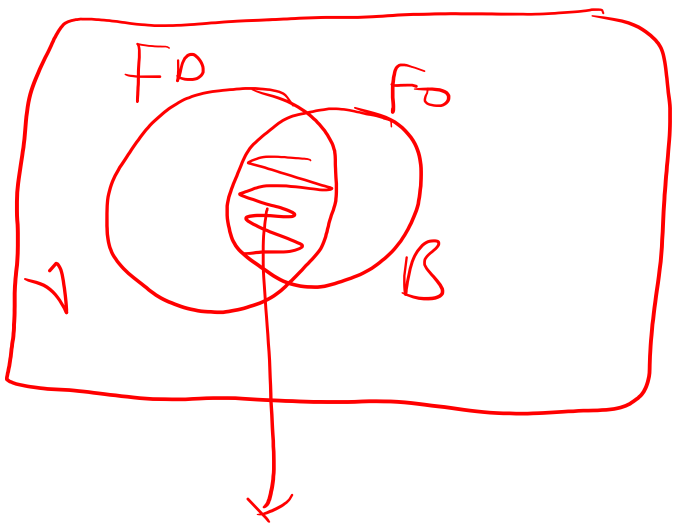
Naïve Bayes Classifier -

P(flight delay given fog) = P (A n B) / P(B)



More the overlap, more the occurrences of flight delay and fog

P(A| B) = P(A n B) / P(B) → eq 1



$$P(A \cap B) = P(B \cap A)$$

$$P(B \cap A) = P(B|A) \times P(A) \quad — ③$$

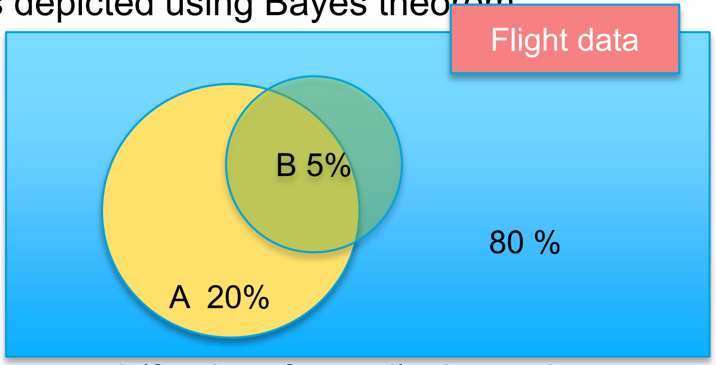
$$P(A|B) = P(A \cap B) \,/\, P(B) \quad — ①$$

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

Naïve Bayes Classifier -

Joint Probabilities (Contd…) -

a. The relationship between dependent events is depicted using Bayes theorem



Likelihood　　Prior prob

Posterior

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

Evidence

b. Probability of event A given that event B has occurred (fog has formed) depends on
   I. Apriori probability of fog occurring whenever there was flight delay – P (B/A)
   II. Apriori probability of flight delay P(A) which is 20% in the example
   III. Apriori probability of flight facing fog P(B) which is 5% in the example

c. When it is a matter of deciding the class of an output such as whether flight will get delayed or not, we calculate P(A/B) and P(!A/B), compare which is higher. Since in both the denominator is P(B), it is ignored as it has no influence on which class will it be

d. However, to calculate the updated probability of a class, denominator P(B) is required

Naïve Bayes Classifier -

a. The following two tables reflect the apriori probabilities of the events A and B. Probabilities based on past data of 100 points

T1	FOG		
Frequency	**Yes**	**No**	**Total**
Flight delayed	4	16	20
Not Delayed	1	79	80
Total	5	95	100

T2	FOG		
Likelihood	**Yes**	**No**	**Total**
Flight delayed	4 / 20	16 / 20	20
Not Delayed	1 / 80	79 /80	80
Total	5 / 100	95 / 100	100

b. In the likelihood table (T2) reveals that P(fog = Yes / flight delayed) = 4/20 = .20 indicating that the probability is 20 percent that a flight is delayed given fog

c. P(A∩B) => P(flight delay | fog) = P(fog / flight delay) * P(flight delay)

d. P(flight delay | fog) = ((4/20) * (20 / 100)) = .04 (maximal probability) (no need to divide by P(B), probability of fog, as it is a constant. **This is Naïve Bayes probability.**

e. **Naïve probability** is when the event of flight delay and fog were unrelated (false independence) P(A∩B) = ((20 / 100) * (5/100)) = .01 This indicates importance of Bayes theorem

Naïve Bayes Classifier -

Suppose there are multiple factors that could lead to flight delay (as shown in the likelihood table below)

T2	FOG		Technical Snag		Pilot Fatigue		Passenger		
Likelihood	**Yes**	**No**	**Yes**	**No**	**Yes**	**No**	**Yes**	**No**	**Total**
Flight delayed	4 / 20	16 / 20	10/20	10/20	0/20	20/20	12/20	8/20	20
Not Delayed	1 / 80	79 /80	14/80	66 /80	8/80	71/80	23/80	57/80	80
Total	5 / 100	95 / 100	24/100	76/100	8/100	91/100	35/100	65/100	100

a. Probability that the flight will be delayed, given that Fog = Yes, Technical Snag = No, Pilot Fatigue = No, Passenger related = Yes is given as –

P(flight delay / Fog **n** ! Technical Snag **n** ! Pilot Fatigue **n** Passenger delay) = (P(Fog **n** ! Technical Snag **n** ! Pilot Fatigue **n** Passenger delay) / flight delay) * P(flight delay)) / P(Fog **n** ! Technical Snag **n** ! Pilot Fatigue **n** Passenger delay)

P(Fog **n** ! Technical Snag **n** ! Pilot Fatigue **n** Passenger delay) / flight delay) **=** P(Fog / flight delay) * P(!Technical snag |flight delay * P(!Pilot Fatigue | flight delay) * P(Passenger Delay| flight delay)

Naïve Bayes Classifier -

Posterior probability – Bayes' rule can be expressed as

$$\boxed{\text{posterior probability} = \frac{\text{conditional probability} \cdot \text{prior probability}}{\text{evidence}}}$$

The posterior probability, in the context of a classification problem, can be interpreted as: "What is the probability that a particular object belongs to class given its observed feature values?"

Naïve Bayes Classifier -

Posterior probability – general expression

$$\boxed{\text{posterior probability} = \frac{\text{conditional probability} \cdot \text{prior probability}}{\text{evidence}}}$$

$$P(\omega_j \mid \mathbf{x}_i) = \frac{P(\mathbf{x}_i \mid \omega_j) \cdot P(\omega_j)}{P(\mathbf{x}_i)}$$

- $\mathbf{x}_i$ be the feature vector of sample i, $i \in \{1, 2, ..., n\}$,
- ω_j be the notation of class j, $j \in \{1, 2, ..., m\}$,
- and $P(\mathbf{x}_i \mid \omega_j)$ be the probability of observing sample $\mathbf{x}_i$ given that is belongs to class ω_j.

Naïve Bayes Classifier -

The objective function is to maximize the posterior probability given the training data

$$\text{predicted class label} \leftarrow \underset{j=1\ldots,m}{\arg\max}\ P(\omega_j \mid \mathbf{x}_i)$$

person has diabetes if P(diabetes | xi) ≥ P(not-diabetes | xi), else classify person as healthy.

Naïve Bayes Classifier (Assumptions) -

One assumption that Bayes classifiers make is that the samples are independent and identically distributed. Samples are drawn from a similar probability distribution.

Independence means that the probability of one observation does not affect the probability of another observation (e.g., time series and network graphs are not independent)

An additional assumption of naive Bayes classifiers is the conditional independence of features. Under this naive assumption, the class-conditional probabilities or (likelihoods) of the samples can be directly estimated from the training data instead of evaluating all possibilities of x

Thus, given a d-dimensional feature vector x, the class conditional probability can be calculated as follows:

$$P(\mathbf{x} \mid \omega_j) = P(x_1 \mid \omega_j) \cdot P(x_2 \mid \omega_j) \cdot \ldots \cdot P(x_d \mid \omega_j) = \prod_{k=1}^{d} P(x_k \mid \omega_j)$$

Naïve Bayes Classifier -

Advantages -

1. Simple , Fast in processing and effective
2. Does well with noisy data and missing data
3. Requires few examples for training (assuming the data set is a true representative of the population)
4. Easy to obtain estimated probability for a prediction

Dis-advantages -

1. Relies on and often incorrect assumption of independent features
2. Not ideal for data sets with large number of numerical attributes
3. Estimated probabilities are less reliable in practice than predicted classes
4. If rare predictor value is not captured in the ** training set but appears in the test set the probability calculation will be incorrect

** For e.g. input record has fog="yes", Technical snag = "yes", Pilot Fatigue = "Yes" and passenger delay = "yes" .If this combination is not in the training set for delayed flights in the past then the probability calculation in step "a" on previous slide will become 0!

Naïve Bayes Classifier -

Lab- 4 Model to predict diabetes among Pima Indians

Description – Sample data is available at
https://archive.ics.uci.edu/ml/datasets/Adult

The dataset has 9 attributes listed below

1. Number of times pregnant
2. Plasma glucose concentration a 2 hours in an oral glucose tolerance test
3. Diastolic blood pressure (mm Hg)
4. Triceps skin fold thickness (mm)
5. 2-Hour serum insulin (mu U/ml)
6. Body mass index (weight in kg/(height in m)^2)
7. Diabetes pedigree function
8. Age (years)
9. Class variable (0 or 1)

Sol: Naive+Bayesian+Pima+Diabetes+.ipynb